# SECTION 9 – PREINSTALLATION CHECKLIST

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## 9–1 INTRODUCTION

"Preinstallation" refers to work necessary to plan and prepare a site for delivery and installation of equipment. Delay, confusion, and waste of manpower can be avoided by completing preinstallation work.

The purpose of this section is to outline key areas of concern in the preparation of a customer's site for Signa Horizon upgrade. It is intended as a guide for GE's Installation Specialists and/or Installation Leaders when making on–site inspections during the installation planning phase of the upgrade. Note site inspections by GE are intended to aid the overall site preparation process. They do not relieve the customer and project architect from responsibility for the design, engineering, and coordination efforts necessary to ensure a successful MR project.

During the course of the site preparation process, GE's Installation Specialists and/or Installation Leaders may observe that the Requested Arrival Date (RAD) for the upgrade equipment is not realistic. If this is the case, appropriate actions must be taken by the GE Field Sales/Service Team to adjust the RAD accordingly.

All work must be in compliance with national and local safety codes.

## 9–2 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE DELIVERY

The following items must be completed by the Installation Specialist or Site Field Engineer prior to upgrade equipment delivery. An on–site inspection by a GE Representative to confirm site readiness must be completed **at least 8 weeks prior to the Scheduled Arrival Date (SAD)** of the upgrade equipment.

☐ 1. Does the system to be upgraded have an LCC Magnet installed?
If yes, the magnet does not need to be upgraded.
If no, has site been planned to meet the specifications for an LCC Magnet, refer to preceding sections of this document?

☐ 2. Does system being upgraded have a Full–size PDU and have arrangements been made for old PDU removal? Refer to Section **2–6–1 Power Distribution Unit (PDU).**

☐ 3. Has a site equipment plan been completed, reviewed with the customer, and finalized?

☐ 4. Has the vibration tests/study been completed per **Section 4–13, VIBRATION**? It is the customer's responsibility to contract a vibration consultant or qualified engineer to implement design modifications to meet the specified limits.

☐ 5. If there were vibration issues; has local GE Field Service verified, with the vibration consultant present, the elimination/reduction of all identified sources?

☐ 6. Has a cable routing plan been completed? Are all cable lengths sufficient? If lengths are not sufficient, a Special Order Inquiry (SOI) will be required. The GE Installation Specialist or Site Field Engineer must submit a SOI Request Form at least 8 weeks prior to SAD to allow for proper processing of the request.
The SOI Request Form is available on the Web at:
http://medquickplace02.ge.com/QuickPlace/mr/PageLibrary862569F30075E524.nsf?OpenDatabase

☐ 7. Does incoming power available at PDU meet the requirements in **Section 5 – POWER REQUIREMENTS**, of this document? This includes correctly sized facility transformer, feeder wire size, Main Disconnect Panel (MDP) wire sizes and circuit breaker rating, and system ground requirements.

## 9–2 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE DELIVERY (Continued)

☐ 8. If GE TwinSpeed Main Disconnect Panel (MDP) is not being used then are plans in place for customer provided MDP to be installed and operational?

☐ 9. Have local and national codes been check to determine if an electrician will be required to provide and/or install new feeder wires to the PDU if ACGD Upgrade is planned? If required then have arrangements been made for an electrician?

☐ 10. Does the site's HVAC (heating, ventilation, and air conditioning) system meet requirements in Section **4–2 TEMPERATURE AND HUMIDITY SPECIFICATIONS** and Section **4–3 AIR COOLING REQUIREMENTS** of this document? If these specifications are not strictly adhered to, failure of the ACGD modules in the ACGD/PDU Cabinet may occur.

☐ 11. Has the site's vibration environment been assessed including system stability tests performed as defined in Section **4–13 VIBRATION**?

☐ 12. Has delivery route been defined for equipment especially enclosure covers(no magnet upgrade) or LCC Magnet with covers if magnet is being upgraded (refer to Section **2–3, MINIMUM DOOR SIZES & DELIVERY ROUTE WEIGHT CAPACITY**) including plans for protecting flooring for heavy equipment & carts (ie. Gradient Body Coil & installation cart)?

☐ 13. Have all upgrade exceptions, such as special cabling and options unique to the site including customer equipment, been identified and clearly tagged **DO NOT REMOVE** ?

☐ 14. For system with Spectroscopy option: Has the site Field Engineer removed the Spectral Module from the Exciter Board in the Systems Cabinet and is the module available at site for re–installation during upgrade.

☐ 15. Has a holding area for old equipment and installation debris been agreed upon with the customer? Making provision for a holding area will allow for efficient unloading and reloading of the delivery vehicle.

☐ 16. Has RF Shielded Room been checked to meet specifications in **Section 7 – RF SHIELDED ROOM** ?

☐ 17. If an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade and the RF Shield is opened, modified, upgraded, etc. then are there plans to recertify (test) the RF Shielded Room integrity after magnet installation to meet or exceed the new system RF Shielded Room specifications. Refer to ***Direction 2294003 Signa TwinSpeed Pre–Installation*** **Section 7–1 RF SHIELDED ROOM SPECIFICATION**.

☐ 18. If the upgrade requires installation of the TRM coil into the magnet at site then have arrangements been made with riggers and lifting equipment equipment to transfer TRM from the shipping cradle to the installation cradle/cart? Refer to **Section 2–3–1 Gradient Coil Assembly, Cradle, & Cart Requirements**.

☐ 19. If the upgrade requires installation of the TRM coil into the magnet at site then have arrangements been made for delivery of the following tools?
- Empty cart/cradle (2134810 / 2134810–2)
  BRM Insertion Tool (2164744–3)
- Ring Alignment Tool (2299496)
- Coil lift brackets (TBD)

☐ 20. If an upgrade to an LCC Magnet is being performed then the cryogenic venting must comply with the specifications in ***Direction 2294003 Signa 1.5T TwinSpeed Pre–Installation*** **Section 4–9 CRYOGENIC VENTING**. The customer is to supply a copy of the Vent Inspection Report to GE Medical Systems for the customer's record file stating compliance to this section.

## 9–3 GENERAL PREINSTALLATION REMINDERS

☐ 1. Has the customer's architect/contractor fully reviewed final site installation drawings using the guidelines provided by MR Siting and Shielding (MRSS)?

☐ 2. Have vehicle parking arrangements been made for installation personnel?

☐ 3. Is temporary storage space available for use during installation?

☐ 4. What is hospital smoking policy?

☐ 5. Are a first aid kit and non–ferrous fire extinguishers available at site?

☐ 6. Have facility arrangements been made for refuse disposal during installation?

## 9–4 SAFETY

**WARNING!**

**THE FOLLOWING SAFETY ITEMS MUST BE ADHERED TO PRIOR TO MOVING THE MAGNET FROM THE TRUCK INTO THE ROOM. THE MAGNET WILL BE EXHAUSTING HELIUM GAS DURING THIS TIME. GASEOUS HELIUM IS AN INVISIBLE, ODORLESS GAS THAT CAN CAUSE ASPHYXIATION WHEN DEPLETING THE AMBIENT OXYGEN SUPPLY. HELIUM GAS IS LIGHTER THAN AIR AND WILL RISE TO THE CEILING.**

☐ 1. Have all of the required room ventilation items been installed and tested to make sure sufficient ventilation is available? Is exhaust fan and fan control installed and functional? The customer is to supply a copy of the Exhaust Fan Test Report to GE Medical Systems for the customers record file. Refer to **Section 4–8 ROOM VENTILATION** for specific requirements.

☐ 2. Are a first aid kit and non–ferrous fire extinguishers available at site?

☐ 3. Are removable chain or strap available for securing cryogen gas cylinders in an upright position to prevent the cylinders from falling which may cause injury or damage?

☐ 4. Are functioning telephones as defined in **Section 2–6–6 System Monitoring & Support Connectivity Requirements** available at site for duration of installation? Phone lines are needed to dial out in case of an emergency.

☐ 5. Has the cryogenic vent been installed complete from the Magnet Room RF shield penetration to the final exit and been inspected to final exit outside of the building? The customer is to supply a copy of the Vent Inspection Report to GE Medical Systems for the customers record file. Refer to **Section 4–9, CRYOGENIC VENTING**.

☐ 6. Have plans been made to connect the cryogenic vent to the magnet after magnet installation in the room? Refer to **Section 4–9, CRYOGENIC VENTING**.

☐ 7. Is there adequate ventilation for helium dewar storage area. Refer to **Section 4–8, ROOM VENTILATION**.

## 9–5 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE 1.5T LCC MAGNET DELIVERY AND MOVING INTO MAGNET ROOM

The following items must be completed prior to magnet delivery. A site inspection by GE Service Representative must be completed prior to magnet delivery to ensure site readiness.

☐ 1. Has Main DIsconnect Panel been installed, electrician wiring complete, MDP operational with power available? Refer to **Table 5–1 REQUIRED CUSTOMER POWER.**

☐ 2. **Is 24 hour/day, 7 days/week power and air cooling available for Main Disconnect Panel (MDP) for powering the TSCC or customer water cooling equipment for Shield /Cryo Cooler (if TGWC to be used), Shield/Cryo Cooler Compressor cabinet, and Magnet Monitor equipment?**

☐ 3. **If MR system configuration is utilizing water cooled TSCC then is 24 hour/day, 7 days/week water cooling available?**

☐ 4. Are all walls and ceiling in magnet room essentially complete except for removable section?

☐ 5. Has a clear route to the magnet room been defined for magnet installation (refer to **Section 2–4, MINIMUM DOOR SIZES**)?

☐ 6. Is a secure space available to store equipment on site?

☐ 7. Have arrangements been made for the use of special rigging equipment for moving the magnet into the magnet room?

☐ 8. Has the raised floor been installed in the equipment/computer room in preparation for the superconducting and resistive power supplies?

☐ 9. Has work in the magnet room been completed or suspended and the magnet room closed off to provide a dust–free, closed environment?

☐ 10. Has all equipment been removed from the magnet room to allow space for the magnet with rigging equipment and cryogen dewars?

☐ 11. Are equipment anchors installed in floor as required for magnet room equipment? Refer to **Section 7–7 ANCHOR SPECIFICATION FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM.**

☐ 12. For all magnet room anchors, has pull test been performed and installation torque value defined? Test results recorded by RF Shield Room Vendor and information forwarded to GE Medical Systems for the customers record file? Refer to ***Direction 2294003 Signa TwinSpeed Pre–Installation*** **Section 7–7–1 Clamping Force And Pull Test.**

☐ 13. For all magnet room anchors, has ground impedance test been performed? Test results recorded by RF Shield Room Vendor and documentation given to GE Medical Systems for the customers record file? Refer to ***Direction 2294003 Signa TwinSpeed Pre–Installation*** **Section 7–7–3 Electrical Isolation.**

## 9–6 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE 1.5T LCC MAGNET CRYOGEN TRANSFER/MAGNET FILL

☐ 1. Have arrangements been made for cryogen delivery and storage?

☐ 2. Does a clear path exist to magnet room for delivery of cryogens?

☐ 3. Is the magnet room adequately vented to allow exhaust of cryogen boil–off?

☐ 4. Is the magnet room clear to provide room for cryogen dewars during filling?

## 9–7 PREPARATIONS REQUIRED IN ADVANCE OF UPGRADE 1.5T LCC MAGNET RAMP–UP

☐ 1. Is the cryogenic vent connected to the magnet?

☐ 2. Has the magnet room been completely closed (removable section closed up and sealed)?

☐ 3. Has the magnet room been tested to ensure that the RF shielding meets the electrical isolation and the100dB requirement for electric/plane waves and the 90dB requirement for magnetic waves refer to **Section 7–1 RF SHIELDED ROOM INTRODUCTION**? The customer is to supply a copy of RF shielded room vendor test reports.

☐ 4. Have all ferrous metal objects been removed from the magnet room?

☐ 3. Have adequate signs (Safety and Exclusion Zones) been posted to warn personnel about dangers of magnetic field?

☐ 4. Have hospital personnel been informed of magnet safety precautions and procedures?

☐ 7. Has power been connected form the Main Disconnect Panel to Power Distribution Unit?

☐ 8. Has the penetration panel been installed?

☐ 9. Is all contractor construction work completed?

☐ 10. Has all contractor equipment that could affect shimming been removed from within the three gauss zone?

☐ 11. Have precautions been taken to prevent movement of large metal objects within the three gauss zone?

☐ 12. Has outside emergency personnel (e.g. fire department and other outside ambulance or emergency personnel been informed of unique characteristics (e.g. strong magnetic field, cryogens, etc.) of magnet and correct precautions to take in event of emergencies?